A Portable Dynamic Student Dropout Early Warning Model

MBAI 5600G

Applied Integrative Analytics Capstone Project - Final Report

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Abstract

This project builds one fixed risk model for students that can be adapted to use different local field names via documented semantic adapters. The main risk model uses 16 numeric features: four required background and timing fields, five optional background fields, four assessment fields and three platform-neutral activity fields. UCI provides snapshot enrolment data while OULAD provides enrolment and snapshots at days 35, 60 and 75. Then XGBoost model is trained on pooled training data and applied to each of the snapshots in exactly the same way: no changes are made to ordering of input features or fitted model parameters. On data that was held out, ROC-AUC values were 0.771 for UCI enrolment snapshot and 0.595, 0.677, 0.694 and 0.689 for enrolment, day 35, day 60 and day 75 respectively. At ROC-AUC values from the last three snapshots of OULAD models, OULAD-only models had increases of 0.009 to 0.013 compared to pooled model. Statistically significant increases were seen for ROC-AUC but not for AP (average precision). An additional model specific to OULAD using 51 numeric features produced a ROC-AUC score of 0.717 for day-35 snapshot. This result shows that ignoring platform-specific details when creating reusable risk models and mapping processes comes at a cost. Results do not prove universal transfer and external validation at another third institution is still needed.

Keywords: student dropout; early warning; model portability; semantic adapter; XGBoost; learning analytics

1. Introduction

Withdrawal from a course is usually the result of a series of things like missing assignments, poor grades, or low attendance. An important business question is whether there is a subset of students (based on combining certain attributes) that advisors can identify and support during enrolment so that withdrawal can be avoided. Models that just predict withdrawal based on final grades perform reasonably well, but advisors will probably not have time to act by that point.

Another issue arises due to portability. UCI student-dropout dataset and Open University Learning Analytics Dataset (OULAD) contain very different tables. For UCI we have admissions records, financial records and semester records. OULAD has registration records, dated assessments associated with modules and learning-platform events [3], [4]. Therefore a model relying solely on raw tables cannot be easily transferred to another school without further work.

Instead this project considers field mapping as part of overall model design. We assume each school maintains its own internal systems and provides adapters that calculate necessary concepts to enable the model to make predictions. Optional feeds may also be empty. Finally output scores will be local rankings for human evaluation only and will not constitute automated withdrawal decisions or calibrated individual probabilities.

2. Problem Statement

Can we develop one predictive model that reliably forecasts student behavior (such as withdrawal) consistently across different institutions where feature columns differ? If yes, what are the main challenges in developing such a model?

Specifically, here are the research questions:

RQ1. Can we reproduce the original experiment from the base paper sufficiently well to set a baseline?

RQ2. Can records from both UCI and OULAD be transformed into a fixed semantic representation so that they score equally using the same trained model?

RQ3. How does a pooled model trained on available data compare to local models at enrolment and later snapshots OULAD?

RQ4. Which features become most important as the availability of dated assessment and activity information increases?

RQ5. What validation steps would another institution need to perform before integrating rankings into an advising workflow?

All experiments were retrospective and used public datasets that had been de-identified. Ontario Tech students did not receive active intervention. Due to resource limitations (all experiments were run on a Mac laptop), we explored predictive models only using logistic regression and specific configurations of XGBoost parameters. Limitations did not force us to limit record numbers.

3. Background & Literature Review

Islam et al. [1] combined academic-prediction techniques with explainable AI and proposed a support system. Using data from UCI they chose as our base paper. Realinho et al. [2] also noted that UCI contains admission variables as well as later-semester measures of outcomes. Grades from first and second semesters are not early-day predictors. We chose to reproduce the original experiment (three-class classification using full features) mainly for comparison. Portable models use only information elements that are relevant to snapshots they consider.

OULAD contains 32,593 records of student modules and more than ten million events on learning platform [4]. Previous research has shown that histories of assessment and engagement of students contribute positively to early-warning predictions [6], [13]. Performance has also been shown in other research to depend heavily on factors such as when predictions were made during term or year, cohorts considered, how outcomes were defined and techniques used for splitting rows in datasets. Row splitting randomly at row level causes problems because students in OULAD can participate in multiple modules and thus appear more than once.

Literature has produced strong models based on single datasets; therefore, our main contribution is not a new classifier. Rather, it is portable measurement layer: one actual model object with fixed semantic input contract; tolerance for missing feed values; paired methods for uncertainty testing; and an example of new school scores. SHAP is used to describe our model [8], but we do not interpret this as causal explanation. The reporting and risk of bias (ROB) frameworks provided by TRIPOD+AI and PROBAST+AI serve as a reference for reporting and bias evaluation, whereas there are no standards for an "educational" use of either framework [9], [10].

4. Data Description

Table 1 describes the two data sources and the role each source serves. There are 4,424 student records available from UCI student-dropout dataset that include a binary indicator of whether the student dropped out for use in cross-platform evaluations. Open University Learning Analytics Dataset (OULAD) provides additional information on registration dates, assessment dates and activity histories. Withdrawal is indicated as positive when the student's unregistration date occurs after the snapshot time. If a student has already withdrawn before the snapshot time, he/she will not appear in subsequent risk sets.

Table 1. Data sources and analytical roles

Dataset	Setting	Main size	Role	Positive outcome
UCI	Portuguese higher education	4,424 students	Base-paper benchmark and enrolment snapshots	Dropout
OULAD	UK distance learning	32,593 student-module records; >10M events	Enrolment and day 35/60/75 snapshots	Future withdrawal after snapshot

In addition, Table 1 illustrates that although the endpoints are related to some extent, they are also distinct. In other words, UCI predicts the final status of a student based upon his/her characteristics; however, OULAD predicts which students will withdraw from a specific module presentation based upon their history up until the current point in time. Rather than eliminating this distinction, we choose to acknowledge it as a limitation.

The size of the risk sets for OULAD changes over time. The risk sets are smaller as time passes since students who have already withdrawn are eliminated. As such, the prevalence decreases from 0.237 at enrolment to 0.134 by day 75. Therefore, average precision should be evaluated in conjunction with prevalence rather than being treated as having comparable class distributions across all cohorts.

Table 2. OULAD snapshot cohorts before model evaluation

Snapshot	Rows	Unique students	Future-withdrawal prevalence
Day 0	29,161	26,085	0.237
Day 35	27,129	24,567	0.174
Day 60	26,253	23,899	0.146
Day 75	25,795	23,566	0.131

4.1 Methodology

Figure 1 illustrates the workflow used to evaluate students from enrolment through the later snapshot days. Student attributes are not inputted into the model directly. Each attribute is first converted by a local adapter into 16 numeric representations. These representations are then fed through a frozen pipeline that includes an imputer that has been fit using training data and an XGBoost model. The output of the model is transformed into percentiles relative to students in the same course and at the same point in time. Advisors may then develop review lists using these locally defined percentiles.

Figure 1. Portable model training and new-school scoring workflow

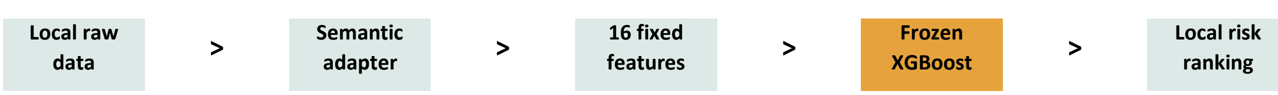

Figure 1 also resolves the issue regarding differences in field names between the various columns of data. The model cannot automatically identify unknown fields. It requires documentation from schools of the meaning of any local fields, including their units of measurement and timing. Additionally, schools must provide documentation explaining how the local fields map to the contract. Although a JSON file example is supplied to handle issues associated with local field names and scaling without requiring a refit to the model, invalid required values, binary fields and course length are rejected.

There are four required features: scaled age, prior education level, study load, and course progress ratio. The remaining twelve feature fields are optional. Random masking was performed during training for an entire group of optional background, assessment and activity features. This enables a school to maintain a valid column order even if it fails to supply one of these types of feed. Only median imputed values based solely on the training data are used.

A partitioning process was used where students were randomly divided into three groups: 60% for training, 20% for validation, and 20% for testing purposes. All modules and points-in-time associated with a given student follow the same partitioning rule. Overlap checks produced zero results. Weights were initially implemented to prevent students with large numbers of module snapshots from overwhelming the model. Second, a weighting process was performed where the sum of weights equals four, one for each institution-by-outcome cell. This was done to ensure that both institutions and both outcome classes are equally weighted in the combined loss function.

4.2 Exploratory Data Analysis

Figure 2 illustrates the original outcome categories from the UCI dataset. Graduate was the most frequent category, with Dropout being the second most frequent. In the portable binary prediction task, Graduate and Enrolled were grouped together to form one non-withdrawal category. However, it is somewhat ambiguous whether an enrolled student had been successful in completing their degree or not since they have simply not dropped out yet.

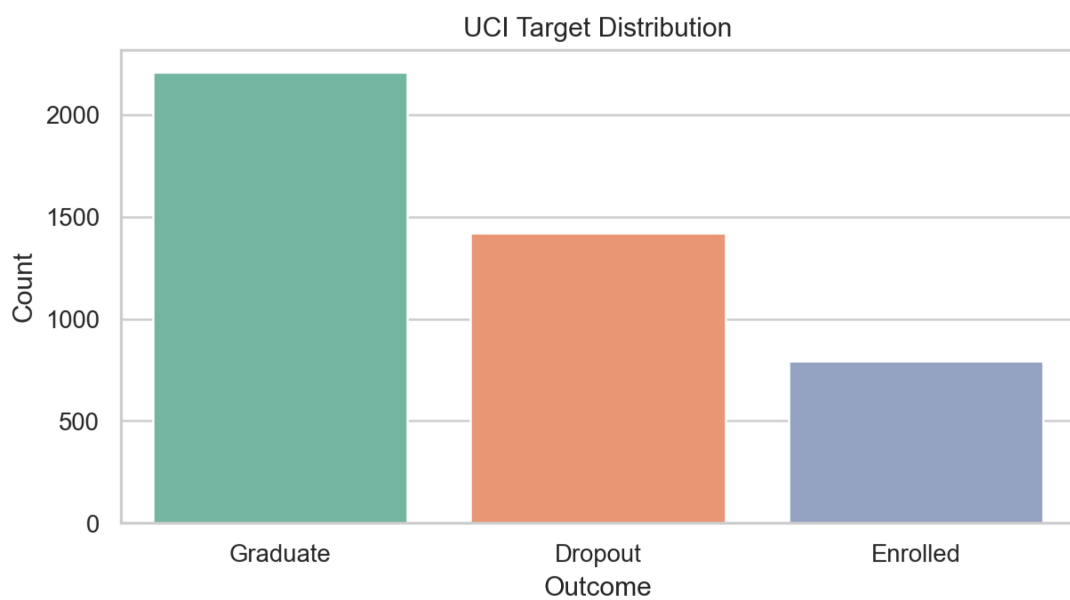

Figure 2. UCI outcome distribution

Figure 3 illustrates the four final-result categories used in OULAD. Students who passed were the most numerous; however, the number of Withdrawn students was less than the total number of students in the Pass and Distinction categories. Since there is no balance among the categories of outcomes, using accuracy alone may provide a distorted view of how well a model performs. An operational target is defined to be more narrow in its definition of what is considered a positive case than withdrawal, because only those students who are currently enrolled in a course and withdraw after a snapshot date are counted as such.

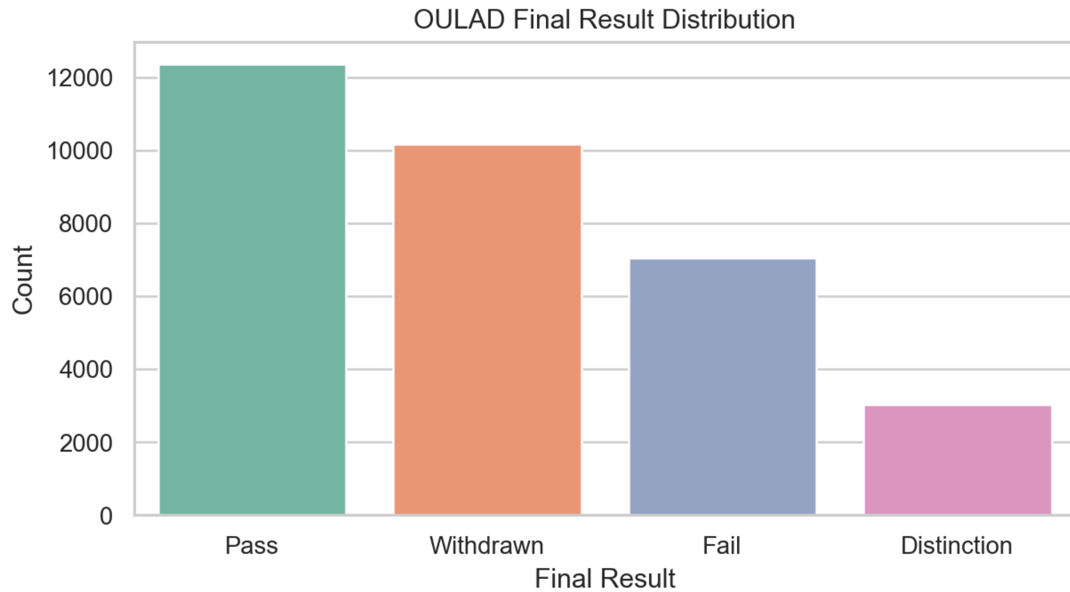

Figure 3. OULAD final-result distribution

Figure 4 shows differences in levels of learning-platform activity among the various categories of final results. Generally, students who withdrew showed lower amounts of activity relative to other categories; however, the amount of activity shown by these groups did vary substantially so that low levels of activity cannot definitively be used to conclude that a student will withdraw. The need for activity-based features such as active days, activity recency, etc., led to the decision not to create individual feature types for each platform click type.

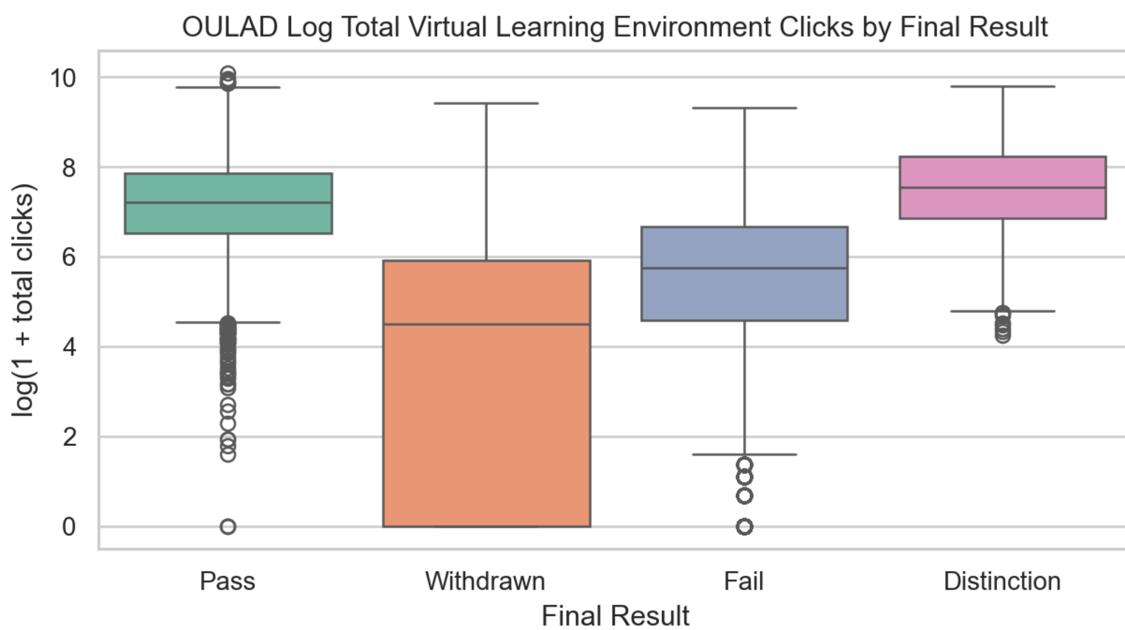

Figure 4. OULAD learning-platform activity by final result

4.3 Data Limitations

UCI does not contain dated learning-platform or assessment events; therefore, it contributes only an enrolment snapshot to the dynamic model. OULAD is the only source that provides information for optional later fields. Thus this research shows that a single model object can support missing or available feature groups; however, it does not demonstrate external validation at the dynamic field level across different institutions. A third educational institution containing dated events will be needed to make such claims.

Approximations occur in mappings at different semantic levels. Financial stability is measured by UCI using debt and tuition status while OULAD measures financial stability based on midpoint deprivation levels. Study load uses denominators specific to the institution. Both institutions have known direction and range but they are not equivalent measurement tools.

5. AI/ML Models

We use logistic regression as a linear baseline and XGBoost as a nonlinear model for this project [7]. Table 3 also lists fixed settings for study of 16 features. Fixed settings were chosen from completed milestones and were kept throughout local, zero-shot and pooled experiments. Thresholds for F1 scores were independently determined using validation data for each snapshot. Operational scoring does not use these thresholds when scoring a new university; rankings are used instead.

Table 3. Main model and preprocessing settings

Component	Setting	Reason
Missing values	Training median; empty columns retained	One fixed schema with optional feeds
Logistic Regression	C=0.2; standardized inputs; 3,000 iterations	Linear baseline
XGBoost	320 trees; depth 5; learning rate 0.03	Nonlinear tabular model
Sampling	subsample=0.85; colsample=0.85	Reduce dependence on individual rows and features
Regularization	min child weight=4; L2=2.0; L1=0.1	Control model complexity
Random seed	42	Reproducible split and training

Accuracy is used exclusively for reporting results when replicating base papers. For binary outcomes precision, recall, F1 score, ROC-AUC and average precision (AP) are reported. Denote counts from the confusion matrix as TP, TN, FP and FN. Denote the classification for record i as y_i and score predicted for record i as p_i and total number of records as n.

$$\text{Accuracy} = (TP + TN) / (TP + TN + FP + FN) \tag{1}$$

$$\text{Precision} = TP / (TP + FP) \tag{2}$$

$$\text{Recall} = TP / (TP + FN) \tag{3}$$

$$F1 = 2 \times \text{Precision} \times \text{Recall} / (\text{Precision} + \text{Recall}) \tag{4}$$

$$\text{ROC-AUC} = P(\text{score}_{positive} > \text{score}_{negative}) \tag{5}$$

$$AP = \sum_k [(\text{Recall}_k - \text{Recall}_{(k-1)}) \times \text{Precision}_k] \tag{6}$$

$$\text{Brier} = (1/n) \times \sum_i (p_i - y_i)^2 \tag{7}$$

Equations (1)-(4) define criteria based on thresholds. Equation (5) defines ranking over positive and negative records. Equation (6) defines average precision in terms of recall. Average precision is useful when classes are imbalanced. Equation (7) defines squared error between a probability prediction and true classification but final deployment score is treated as uncalibrated.

1,000 bootstrap samples of clusters of students and 1,000 paired permutations of those clusters were used for comparison. Records of each student were resampled together or swapped together. Two-sided Monte Carlo p-values were computed according to Equation (8), where B is the number of permutations.

$$p = [1 + \sum_b I(|d_b| >= |d_{observed}|)] / (B + 1) \tag{8}$$

SHAP summaries use mean absolute contribution defined in Equation (9). Mean absolute contribution measures on average how much a feature changes fitted model output; this does not mean altering the feature will change retention.

$$\text{Mean absolute SHAP}_j = (1/n) \times \sum_i |\text{SHAP}_{ij}| \tag{9}$$

6. Experimental Results and Analysis

6.1 Main Portable Dynamic Model

Table 4 shows the primary results for one pooled XGBoost model. UCI enrolment ROC-AUC is 0.771. ROC-AUC for OULAD improved from 0.595 at enrolment to 0.677 at day 35 and 0.694 at day 60 and remained at 0.689 at day 75. Average precision declines after day 35 partly because the prevalence of future withdrawal decreases.

Table 4. One pooled dynamic model across held-out contexts

Held-out context	Rows	Prevalence	ROC-AUC	AP	Precision	Recall	F1
OULAD enrolment	5,811	0.237	0.595	0.310	0.237	1.000	0.384
OULAD day 35	5,408	0.175	0.677	0.311	0.309	0.450	0.367
OULAD day 60	5,240	0.148	0.694	0.292	0.279	0.504	0.359
OULAD day 75	5,154	0.134	0.689	0.270	0.241	0.485	0.322
UCI enrolment	885	0.321	0.771	0.635	0.469	0.827	0.599

Figure 5 also shows ranking and threshold metrics. While the model performs moderately well as a ranking system, raw scores should never be shown as individual dropout probabilities. F1 at day 0 for OULAD represents a validation threshold that identified almost everything positive but also incorrectly identified a lot of negatives and that is another reason to use local capacity review rules.

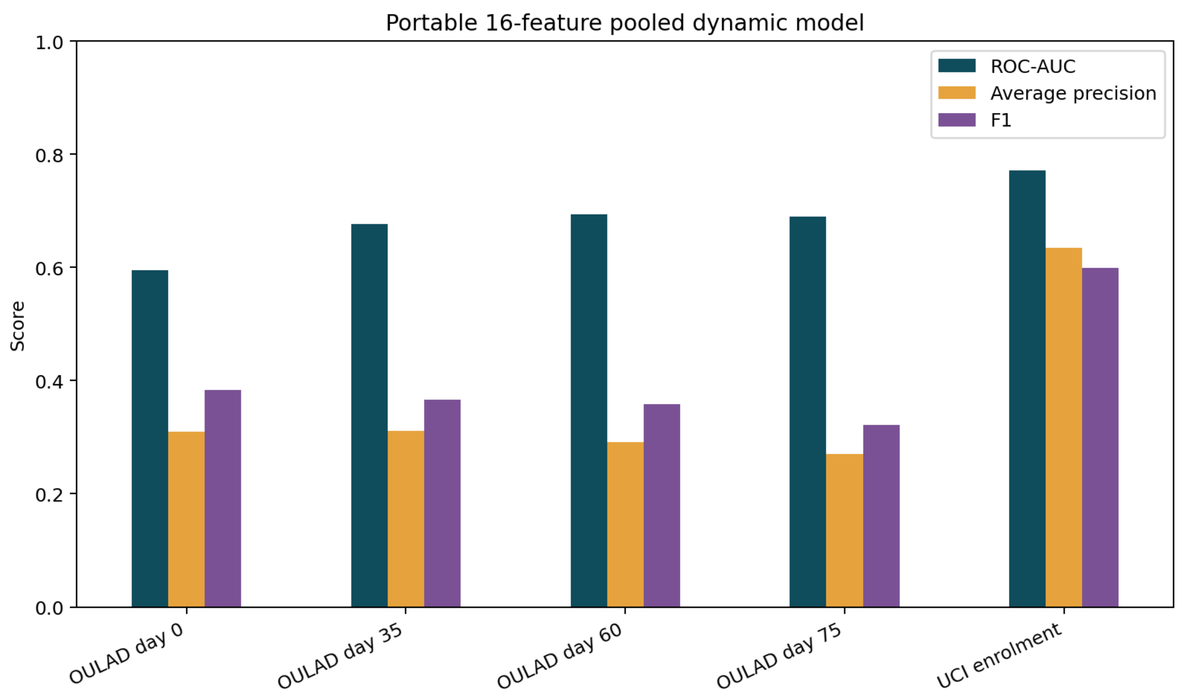

Figure 5. Performance of one pooled 16-feature dynamic model

6.2 Portability Cost and Statistical Testing

Table 5 compares pooled XGBoost versus local-only using identical records that have been held out. At UCI and OULAD enrolment times there were no significant differences. At days 35, 60 and 75 of OULAD, ROC-AUC pooled results were lower by 0.009, 0.013 and 0.009 respectively. Paired differences were significant but quite small; AP differences were not significant.

Table 5. Paired student-cluster ROC-AUC tests for pooled minus local-only XGBoost

Comparison	Difference	95% cluster CI	p-value	Significant
UCI day 0: pooled minus local-only	-0.003	[-0.011, +0.007]	0.561	No
OULAD day 0: pooled minus local-only	-0.001	[-0.008, +0.006]	0.740	No
OULAD day 35: pooled minus local-only	-0.009	[-0.015, -0.003]	0.005	Yes
OULAD day 60: pooled minus local-only	-0.013	[-0.018, -0.007]	0.001	Yes
OULAD day 75: pooled minus local-only	-0.009	[-0.015, -0.003]	0.005	Yes

Table 5 shows conclusions from revised analysis in comparison to conclusions from an older draft. Unified model performed significantly worse later snapshots; its advantage lay in operational reuse with measurable penalty in performance rather than greater predictive accuracy.

Table 6 compares performance of unified model against a benchmark based on 51 features specific to OULAD. Specialized model consistently outperforms at later snapshots and scores 0.717 at day 35. This model uses detailed assignment and platform data which portable schema explicitly removed. Rows in Table 6 do not support paired significance claims because different cohorts were used for the two analyses.

Table 6. Portable dynamic model and OULAD-specific benchmark

Model scope	Day	ROC-AUC	AP	F1
Portable 16	35	0.677	0.311	0.367
Portable 16	60	0.694	0.292	0.359
Portable 16	75	0.689	0.270	0.322
OULAD-specific 51	35	0.717	0.356	0.403
OULAD-specific 51	60	0.718	0.305	0.379
OULAD-specific 51	75	0.717	0.277	0.343

6.3 Explanation Across Snapshots

SHAP values pooled in Figure 6 explain model behavior across different snapshots. Financial stability and age contribute more to UCI enrolment scores. At OULAD day 35, average assessment scores carry greater importance. Because the model is shared across time, the contribution of course progress ratio is larger at later OULAD snapshots. Because progress ratio is constant within a snapshot, it serves as a baseline adjustment for the model rather than ranking students within a snapshot; it can shift the baseline across snapshots but cannot independently rank students within the same snapshot. Therefore Figure 6 shows that this should not be targeted for intervention.

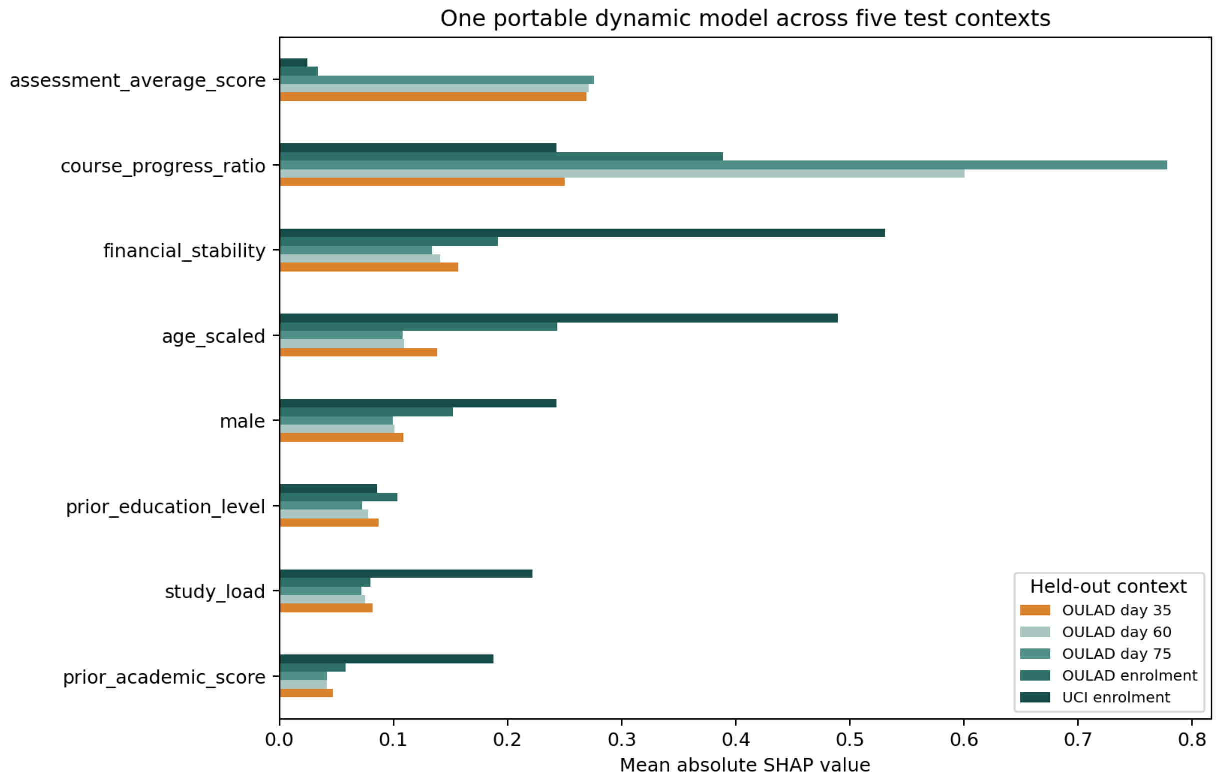

Figure 6. Mean absolute SHAP for one model in five held-out contexts

Local explanation is supported by these feature shifts. Advisors should base any actions on locally relevant data. Low scores might indicate outreach to academics first. Missing activity might first need verification of access, employment or technology issues. Sensitive fields should never serve as basis for adverse action.

6.4 Enrolment-Only Baseline and Supporting Studies

We keep a fallback enrolment model with six features for schools that don't have event histories. Table 7 shows pooled ROC-AUC is 0.777 on UCI and 0.624 on OULAD. Direct transfer from UCI to OULAD produces only ROC-AUC of 0.481. Results support pooled training but they also show that using common feature names doesn't remove institutional differences.

Table 7. Six-feature enrolment model results

Experiment	ROC-AUC	AP
UCI local -> UCI	0.777	0.644
OULAD local -> OULAD	0.635	0.418
UCI -> OULAD zero-shot	0.481	0.298
OULAD -> UCI zero-shot	0.603	0.407
Pooled -> UCI	0.777	0.635
Pooled -> OULAD	0.624	0.409

Figure 7 shows earlier local-adaptation analysis and direct-transfer experiments for enrolment models. These experiments provide supporting evidence rather than representing finalized portable dynamic design. An important finding is that even a small amount of target institution data reveals weaknesses before a model goes operational.

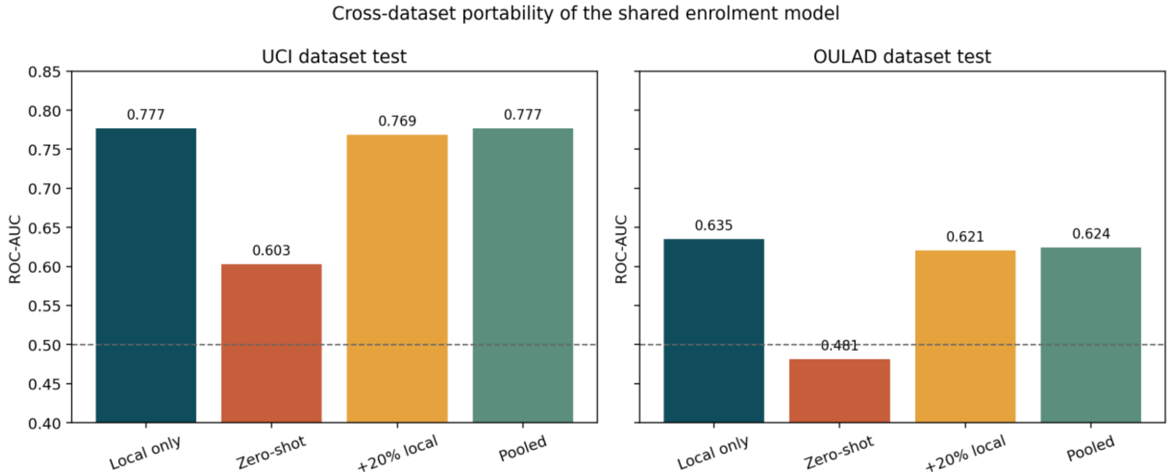

Figure 7. Six-feature local, zero-shot, adapted, and pooled transfer results

Reproducing the base paper got accuracy of 0.768 and macro F1 score of 0.704 on three-class task from UCI. Although this accuracy is lower than Islam et al. [1] reported, reproduction is considered reasonably close because full original protocol was unavailable. Using stricter benchmark UCI day-one binary, XGBoost scored ROC-AUC of 0.832 and F1 score of 0.666. Analysis specific to OULAD produced values of 0.717, 0.718 and 0.717 at days 35, 60 and 75 respectively. When using a presentation from 2014J as holdout data, ROC-AUC drops to 0.678. Results show that performance can vary even within the same dataset depending on time and institutional context.

Figure 8 compares performance of OULAD at different snapshot times. ROC-AUC remained stable while AP declined as cohorts changed over time. When using a controlled cohort at day 75, inclusion of later feature windows also improved rankings of students. Comparison separates effect of additional information from effect of earlier withdrawals on the cohort.

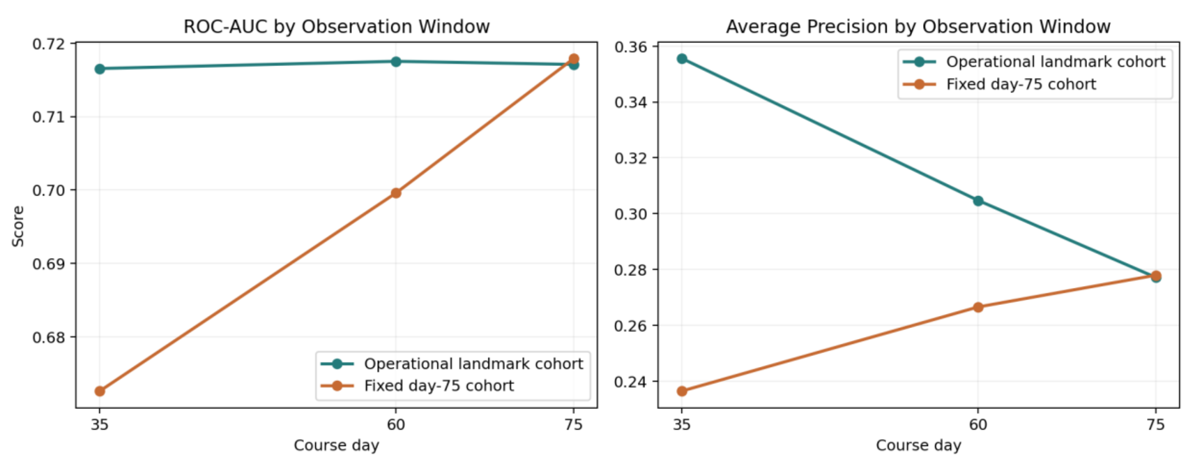

Figure 8. OULAD-specific temporal performance and controlled-cohort comparison

7. Business Impact

The proposed product includes a queue for advisors to review students. Each school decides what percentage of currently enrolled students advisors can review at the same snapshot date. For example, a school might review highest-ranked 15% of students in each course at the same snapshot date. So if there are 1,000 active enrolments, this rule creates a list of 150 students. This does not mean that all 150 students will withdraw or that every student should receive identical support.

Before alerts are sent to staff, the school should run a silent historical first pass. This process checks whether the model produces expected results, whether semantic adapters map correctly and whether alert workflows work as intended. The school also confirms when withdrawals are recorded relative to each snapshot, evaluates performance of rankings and subgroups and verifies scores remain consistent across different runs and versions of models. Separate version numbers should be assigned to both model and semantic adapters so that changes can be tracked.

Table 8. Minimum third-school pilot gates

Gate	Evidence	Decision
Semantic mapping	Document source, unit, range, and availability date	Reject unclear mappings
Technical validation	No missing required fields; expected optional coverage	Repair feed before scoring
Historical shadow test	ROC-AUC, AP, calibration, and capacity precision	Do not use weak rankings
Fairness review	Error and score checks by approved subgroups	Restrict or revise sensitive features
Operational test	Advisor capacity, response time, and referral options	Set local review share
Monitoring	Drift, mapping changes, and outcomes by term	Pause when contract changes

Table 8 lists the minimum requirements for a pilot at another institution. The financial benefits of the system cannot be calculated unless we know how many students who received support stayed enrolled and how many would have stayed or withdrawn without support. Therefore this study cannot claim specific savings or improvement in retention. However using an existing model and tables from the institution itself might require less work to implement.

Further research is needed to determine whether support after alerts improves retention [14]. Follow-up study would need a control group so that we can compare outcomes for students who received support with those who did not. This project ranks students according to risk of withdrawal rather than finding the most effective intervention. Future research should therefore compare different support actions and measure their effects on persistence among students.

8. Ethical and Responsible AI

Student data remain very sensitive even when names are no longer available. Access should be limited to staff who provide support to students. Schools should set retention periods for data, maintain audit logs and provide a process for students to appeal or correct information. Risk scores should not be used to decide admission, take disciplinary action, impose financial penalties or reduce services to students. Guidance from NIST AI Risk Management Framework [11] helps assign responsibility, define ownership and monitor the system.

Characteristics such as age, gender, disability status, and variables related to financial circumstances can reproduce existing inequalities. Gender and declared support needs are optional fields in semantic contract. Schools should evaluate both full model and restricted version and compare error rates and score distributions among different groups of students. SHAP values do not show that a model is fair and they cannot explain why an individual student decides to leave.

Human review must take place before any action is taken. Alerts should start supportive conversations and checking underlying data rather than presenting them as facts about students. Advisors also need access to realistic support services because generating alerts by itself doesn't add value.

9. Limitations and Future Work

- There were only two publicly accessible datasets analyzed for this research; there were no dated dynamic data elements in UCI, while OULAD was the only dataset that provided them.
- Although UCI dropout and OULAD module withdrawal represent similar student outcomes, they are not exact representations of the same student outcome.
- All the semantic mappings developed for this project are based on documented approximations, and have not been formally verified as equivalent measures.
- For the later time points of the OULAD data, the performance of the pooled model is slightly poorer, however, it is still statistically significantly worse than the separate OULAD-only models.
- Neither the threshold values determined by the model, nor the actual raw scores, have been calibrated at any other university.
- This research is retrospective in nature, therefore, the results cannot demonstrate whether an increase in advisor contact with students increases their likelihood of retaining their enrolment status.

Next steps include conducting a pre-registered evaluation at a third university. Prior to examining additional outcomes, all aspects of the 16-feature contract, the saved model, and the code associated with evaluating the model from outside should be frozen. The institution should initially use the semantic mapping contract to map a previously enrolled student cohort. Documenting feature availability, feature coverage, and data drift should occur prior to documenting and validating the unchanged model's ranking performance at each of the established snapshot times. If the initial model does poorly when providing rankings, either limited local adaptation, or developing a hierarchical model could be considered as alternative methods to improve upon the original external testing, without modifying the original external test. In addition to a prospective study comparing different student retention outcomes between students who received a review versus those who did not receive a review.

10. Conclusions

The outcome of this research has been the creation of a single deployable model object, one 16-feature semantic contract, one configurable mapping script, and one scoring script which functions with the absence of optional feed data. The pooled model produced a ROC-AUC score of 0.771 using the UCI enrolment dataset and 0.677 to 0.694 for the subsequent OULAD snapshots. It is logical that the local OULAD models would perform marginally better than the pooled model; as well, the OULAD-specific 51-feature model was also considerably stronger. In conclusion, there exists a trade-off between the portability of a model and its performance.

As such, instead of developing a universally applicable "plug-and-play" predictor, we have developed a reusable model with embedded local measurements, local validations, ranking-based utilization, and human oversight.

11. References

[1] M. M. Islam, F. H. Sojib, M. F. H. Mihad, M. Hasan, and M. Rahman, "The integration of explainable AI in Educational Data Mining for student academic performance prediction and support system," Telematics and Informatics Reports, vol. 18, Art. no. 100203, 2025.

[2] V. Realinho, J. Machado, L. Baptista, and M. V. Martins, "Predicting Student Dropout and Academic Success," Data, vol. 7, no. 11, Art. no. 146, 2022.

[3] V. Realinho et al., "Predict Students' Dropout and Academic Success," UCI Machine Learning Repository, 2021. doi: 10.24432/C5MC89.

[4] J. Kuzilek, M. Hlosta, and Z. Zdrahal, "Open University Learning Analytics dataset," Scientific Data, vol. 4, Art. no. 170171, 2017.

[5] B. Carballo-Mendivil et al., "Predicting Student Dropout from Day One: XGBoost-Based Early Warning System Using Pre-Enrollment Data," Applied Sciences, vol. 15, no. 16, Art. no. 9202, 2025.

[6] A. Shaikhanova et al., "Interpretable Predictive Modeling for Educational Equity," Big Data and Cognitive Computing, vol. 9, no. 11, Art. no. 297, 2025.

[7] T. Chen and C. Guestrin, "XGBoost: A scalable tree boosting system," in Proc. ACM SIGKDD, 2016, pp. 785-794.

[8] S. M. Lundberg and S.-I. Lee, "A unified approach to interpreting model predictions," in Advances in Neural Information Processing Systems, vol. 30, 2017.

[9] G. S. Collins et al., "TRIPOD+AI statement," BMJ, vol. 385, e078378, 2024.

[10] K. G. M. Moons et al., "PROBAST+AI," BMJ, vol. 388, e082505, 2025.

[11] National Institute of Standards and Technology, Artificial Intelligence Risk Management Framework (AI RMF 1.0), NIST AI 100-1, 2023.

[12] C. Romero and S. Ventura, "Educational data mining and learning analytics: An updated survey," WIREs Data Mining and Knowledge Discovery, vol. 10, no. 3, e1355, 2020.

[13] F. Qiu et al., "Predicting students' performance in e-learning using learning process and behaviour data," Scientific Reports, vol. 12, Art. no. 453, 2022.

[14] S. Plak et al., "Early warning systems for more effective student counselling in higher education," Higher Education Quarterly, vol. 76, no. 1, pp. 131-152, 2022.

Appendix A. Portable Dynamic Feature Contract

Table A1 details all fixed model inputs. Each scored record must contain the necessary fields. The frozen pipeline will replace missing or blank optional fields with estimated values. All observed values exist within documented 0-to-1 ranges.

Table A1. Sixteen-feature semantic contract

Feature	Status	Group	Meaning
age_scaled	Required	background	Age position on a fixed 18-60 scale
prior_education_level	Required	background	Highest prior education on a documented 0-1 ladder
study_load	Required	background	Registered load relative to a heavy full load
course_progress_ratio	Required	timing	Share of planned course duration elapsed
prior_academic_score	Optional	background	Normalized pre-entry grade or GPA
previous_attempts	Optional	background	Prior attempts capped and scaled to 0-1
male	Optional	sensitive_optional	Optional documented binary gender indicator
declared_support_need	Optional	sensitive_optional	Optional declared disability or support need
financial_stability	Optional	sensitive_optional	Optional documented financial-stability proxy
assessment_completion_rate	Optional	assessment	Share of due assessment weight submitted by snapshot
assessment_average_score	Optional	assessment	Weighted normalized score among submitted assessments
assessment_score_available	Optional	assessment	Whether at least one score is available at snapshot
late_submission_rate	Optional	assessment	Share of submitted assessments submitted after due date
active_day_rate	Optional	activity	Share of observable course days with LMS activity
days_since_last_activity_scaled	Optional	activity	Days since last LMS activity capped at 30 and divided by 30
recent_activity_rate	Optional	activity	Share of observable days active in the last 14 days

All elements from Table A1 are represented in both the JSON contract and mapping scripts. Course IDs, regions, assignment numbers, and platform activity types were removed due to lack of reliability in cross-environmental applicability.

Appendix B. Reproducibility Checklist

Table B1. Submitted code and data package

Item	Location	Purpose
Main model	code/cross_school_dynamic_model.py	Build snapshots, train, test, explain, and save model
New-school mapper	code/map_new_school_snapshots.py	Convert local columns and units
Scoring	code/score_new_school_dynamic.py	Create ranking, percentile, and review flag
Output checks	code/check_outputs.py	Check schemas, splits, artifacts, metrics, and examples
Frozen artifact	models/unified_dynamic_xgboost.joblib	One fitted pipeline for scoring
Contract	models/unified_dynamic_feature_contract.json	Feature definitions and inference policy
Raw data	data/raw	Packaged UCI and OULAD sources
Results	outputs/results	Metrics, predictions, tests, SHAP, and examples

Table B1 represents the major reproducibility files. The README document includes commands to run the project under both Windows, macOS, and Linux operating systems. The random seed used during training was set at 42. Exact dependency requirements can be found in requirements.txt. check_outputs.py must end with 'Output check passed.'